

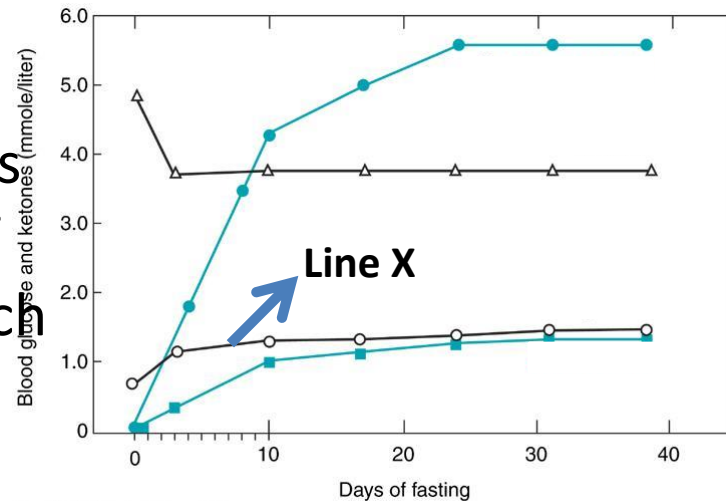
# **DM\_IMCQ 6 - KEY**

1. Under aerobic conditions, in an actively contracting skeletal muscle the malate-aspartate shuttle was active. Which enzyme provides the NADH for this shuttle?

- A. Succinate dehydrogenase 20%
- B. Lactate dehydrogenase 20%
- C. Glyceraldehyde 3-phosphate dehydrogenase 20%
- D.  $\alpha$ -ketoglutarate dehydrogenase complex 20%
- E. Pyruvate dehydrogenase complex 20%



2. The graph represents the changes in blood levels of various compounds on different days of fasting in normal volunteers. Which organ/enzyme activity produces the compound labeled as 'X'?



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- |                                                     |    |
|-----------------------------------------------------|----|
| A. Endothelium/ lipoprotein lipase                  | 0% |
| B. Adipocyte/ hormone sensitive lipase              | 0% |
| C. Liver/ HMG CoA lyase                             | 0% |
| D. Liver/ Glucose 6-phosphatase                     | 0% |
| E. Brain/ Succinyl CoA acetoacetate CoA transferase | 0% |
| F. Intestine/ Pancreatic lipase                     |    |



3. A 5-day-old infant with lethargy presents with the following blood reports. Which of the following gene mutations is most likely?

| Parameter                     | Patient results (Normal) |
|-------------------------------|--------------------------|
| pH                            | 7.2 (7.4)                |
| HCO <sub>3</sub> <sup>-</sup> | 18 (24)                  |
| Blood glucose                 | 100 (80-100)             |
| Lactate                       | Increased                |



- A. Glucose 6-phosphatase 0%
- B. Pyruvate dehydrogenase complex 0%
- C. Medium chain acyl CoA dehydrogenase 0%
- D. Glucokinase 0%
- E. Pyruvate carboxylase 0%

4. Di Abietes, is on treatment with insulin for the management of type I diabetes mellitus for the past 5 years. She has forgotten to take insulin for two days, and has been brought to the ER. Which metabolic process would be active at the time of admission?

- |                                     |    |
|-------------------------------------|----|
| A. Glycogenesis in liver            | 0% |
| B. Glucose uptake by adipose tissue | 0% |
| C. Gluconeogenesis in liver         | 0% |
| D. TAG synthesis in adipose tissue  | 0% |
| E. Glycogenesis in muscle           | 0% |



5. A research student has isolated mitochondria of the hepatocytes. Which enzymes are localized to this organelle?

- A. Oxidation of very long chain fatty acids 0%
- B. Hormone sensitive lipase 0%
- C. Fatty acid biosynthesis 0%
- D. Degradation of glycolipids 0%
- E. Enzymes of ketogenesis 0%
- F. All enzymes of the urea cycle 0%
- G. All enzymes of gluconeogenesis 0%



6. A 6-month-old infant was brought to the hospital with seizures. The following lab reports were obtained on admission. Which of the following conditions is most likely?

|                   | Patient values | Reference ranges |
|-------------------|----------------|------------------|
| Blood glucose     | 45             | 70-90 mg/dL      |
| 3-hydroxybutyrate | 0.3            | 0.4-0.5 mmol/L   |



- A. Refsum disease 0%
- B. Zellweger syndrome 0%
- C. MCAD deficiency 0%
- D. Myopathic carnitine deficiency 0%
- E. CPT-II deficiency 0%
- F. Inherited urea cycle disorder 0%
- G. Glucagon secreting tumor 0%



7. A 14-year-old teenager has history of muscle cramps. Muscle biopsy reveals normal glycogen levels but increased TAG droplets. His blood glucose and ketone body levels were within normal limits. Which of the following is defective?

- A. Oxidation of very long chain fatty acids 0%
- B. Oxidation of branched chain fatty acids 0%
- C. Muscle glycogenolysis 0%
- D. Transport of fatty acids into mitochondria 0%
- E. Endothelial lipoprotein lipase activity 0%
- F. Degradation of sphingomyelin 0%
- G. Targeting of enzymes to the lysosomes 0%
- H. Adipose tissue lipolysis 0%





8. A 10-year-old girl has a sickling crisis.  
Which of these lab findings are likely?

- |                                                                       |                                                                                     |    |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------|----|
| A. Urine bilirubin absent                                             |  | 0% |
| B. Urine urobilinogen absent                                          |                                                                                     | 0% |
| C. Serum direct bilirubin increased                                   |                                                                                     | 0% |
| D. Serum unconjugated bilirubin and<br>conjugated bilirubin increased |                                                                                     | 0% |
| E. Elevated serum ALT and AST                                         |                                                                                     | 0% |
| F. Elevated serum ALP and GGT                                         |                                                                                     | 0% |



9. The following are the lab reports of an 28-year-old male following a brief illness:

Total bilirubin: 3mg/dL (Normal:<1mg/dL)

Direct bilirubin: 0.2 mg/dL (Normal: <0.2 mg/dL)

ALT and AST: Normal; ALP and GGT: Normal

Which of the following is the most likely?



- |                                                     |    |
|-----------------------------------------------------|----|
| A. Acute viral hepatitis                            | 0% |
| B. Chronic alcoholic cirrhosis                      | 0% |
| C. Gall stone obstructing the common bile duct      | 0% |
| D. Partial deficiency of UDP glucuronyl transferase | 0% |
| E. Deficiency of the bilirubin ABC transporter      | 0% |
| F. Crigler-Najjar syndrome I                        | 0% |
| G. Rickets                                          |    |



## 10. Which of the following statements about lipid digestion and absorption is correct?

- A. The formation of chylomicrons takes place in the intestinal lumen 0%
- B. The end product of triacylglycerol digestion is glycerol and free fatty acids 0%
- C. The primary enzyme for dietary lipid digestion is pancreatic lipase in adults 0%
- D. The end products of lipid digestion are present in the mucosal cell as mixed micelles 0%
- E. Bile salts help in the formation of chylomicrons within the mucosal cell 0%



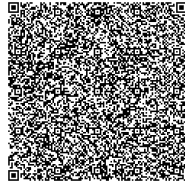
11. In a child with Dubin-Johnson syndrome, which of these lab findings is most likely present?

- |                                                          |    |
|----------------------------------------------------------|----|
| A. Increased urine urobilinogen                          | 0% |
| B. Increased serum indirect bilirubin                    | 0% |
| C. Increased serum ALT                                   | 0% |
| D. Increased serum ALP and GGT                           | 0% |
| E. Increased serum conjugated bilirubin                  | 0% |
| F. Decreased hepatic UDP-glucuronyl transferase activity | 0% |
| G. Decreased serum albumin                               | 0% |



12. Lab reports of a 73 year-old woman are given below. Which condition is most likely?

|                  | Patient values | Reference ranges |
|------------------|----------------|------------------|
| ALT              | 60             | 20-40 U/ L       |
| AST              | 50             | 20-40 U/ L       |
| ALP              | 500            | 60-70 IU/ L      |
| GGT              | 520            | 50-70 U/L        |
| Total bilirubin  | 5              | 0.8-1 mg/ dL     |
| Direct bilirubin | 4.8            | 0.1-0.2 mg/dL    |



- A. Gilbert syndrome 0%
- B. Dubin Johnson syndrome 0%
- C. Sickle cell crisis 0%
- D. Carcinoma of head of pancreas 0%
- E. Bone disease 0%
- F. Chronic alcoholic cirrhosis 0%
- G. Acute viral hepatitis 0%



13. A 25-year-old woman is fasting for 8 hours for estimation of fasting blood glucose. Which of the following hepatic enzymes would be active at this time?

- |                                        |    |
|----------------------------------------|----|
| A. Carnitine palmitoyl transferase - I | 0% |
| B. Glycogen synthase                   | 0% |
| C. Pyruvate kinase                     | 0% |
| D. Phosphofructokinase-1               | 0% |
| E. Glucokinase                         | 0% |
| F. Acetyl CoA carboxylase              | 0% |



14.  $\beta$ -oxidation of a saturated fatty acid in the cardiac muscle produced 8 acetyl CoA. Which of the following fatty acid was oxidized in this process?

- A. Oleic acid 0%
- B. Palmitic acid 0%
- C. Stearic acid 0%
- D. Linoleic acid 0%
- E. Phytanic acid 0%
- F. Arachidonic acid 0%
- G. Linolenic acid 0%



15. A 32-year-old man has destruction of the brush border of intestinal mucosal cells. Which of the following enzymes are present at this location?

- |                     |     |
|---------------------|-----|
| A. Alpha-Amylase    | 14% |
| B. Lipase           | 14% |
| C. Lactase          | 14% |
| D. Isomaltase       | 14% |
| E. Maltase          | 14% |
| F. Trypsin          | 14% |
| G. Phospholipase A2 | 14% |







16. During fasting, palmitoyl CoA inhibits acetylCoA carboxylase by promoting \_\_\_\_\_

- |                      |    |
|----------------------|----|
| A. Polymer formation | 0% |
| B. Phosphorylation   | 0% |
| C. Dephosphorylation | 0% |
| D. Monomer formation | 0% |
| E. Down-regulation   | 0% |



17. In the well-fed state, newly synthesized fatty acids are not metabolized by  $\beta$ -oxidation because of high intracellular levels of

- |                              |    |
|------------------------------|----|
| A. Palmitoyl CoA             | 0% |
| B. Long chain fatty acyl CoA | 0% |
| C. Acetyl CoA                | 0% |
| D. Malonyl CoA               | 0% |
| E. Carnitine                 | 0% |



18. In a patient with myocardial infarction, the cardiac muscle becomes hypoxic. Which of the following provides energy to the cardiac muscle under these conditions?

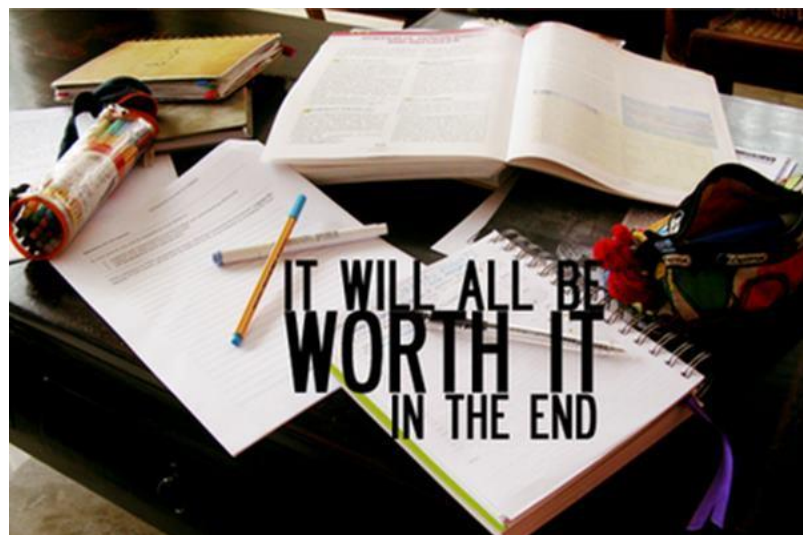
- |                                  |    |
|----------------------------------|----|
| A. Krebs cycle                   | 0% |
| B. Electron transport chain      | 0% |
| C. ATP synthase                  | 0% |
| D. Glycolysis                    | 0% |
| E. PDH complex                   | 0% |
| F. Beta oxidation of fatty acids | 0% |



## Acetyl CoA

```
graph TD; B1[ ] --> AC[Acetyl CoA]; B2[ ] --> AC; B3[ ] --> AC; AC --> B4[ ]; AC --> B5[ ]; AC --> B6[ ]; AC --> B7[ ]
```

1. Write the precursors and various fates of acetyl CoA
2. Identify organs/ organelles where it is active
3. Mention regulated step in each pathway
4. Indicate which pathways are active in fed and fasting state



## Answer Key

1. C
2. B
3. B
4. C
5. E
6. C
7. D
8. A
9. D
10. C
11. E
12. D
13. A
14. B
15. C, D, E
16. D
17. D
18. D